Postiz Social Channel Authentication & .env Reference

Step-by-Step Developer Portal Setup, OAuth Redirect URIs, and Credential Formatting to Resolve "Invalid App ID" Errors

1. Why "Invalid App ID" & Connection Errors Occur in Postiz

According to the Postiz Provider API Keys Overview, Postiz displays every supported social platform in the **Add Channel** dialog regardless of whether its environment variables are configured. Clicking a platform whose keys are missing, empty, or misconfigured triggers an immediate OAuth rejection from the upstream provider (most commonly **"Invalid App ID"** on Meta platforms or failed redirects on X, LinkedIn, and TikTok). To eliminate these errors across your local environment (`FRONTEND_URL="http://localhost:4200"`), verify these five architectural rules:

- **Populate Unconfigured Provider Keys:** In the baseline `.env` file, social provider keys are omitted. When you click Facebook, Instagram, or Threads without setting `FACEBOOK_APP_ID`, `INSTAGRAM_APP_ID`, or `THREADS_APP_ID`, Postiz sends an empty `client_id` parameter in the OAuth query string, causing Meta to return *"Invalid App ID: The provided App ID does not look like a valid App ID."*
- **Use Exact Sub-Product App IDs on Meta (Crucial):** Meta generates **three distinct App IDs** inside a single Meta Developer Application. Using the top-level Meta App ID for Instagram Standalone or Threads will trigger an "Invalid App ID" error:
  - **Facebook Page & Instagram (Facebook Business):** Use the main **App ID** and **App Secret** from *App Settings → Basic* (mapped to `FACEBOOK_APP_ID` and `FACEBOOK_APP_SECRET`).
  - **Instagram (Standalone):** Use the dedicated **Instagram App ID** and **Instagram App Secret** found inside *Products → Instagram → API setup with Instagram business login* (mapped to `INSTAGRAM_APP_ID` and `INSTAGRAM_APP_SECRET`).
  - **Threads:** Use the dedicated **Threads App ID** and **Threads App Secret** found inside *Use Cases → Access the Threads API → Settings* (mapped to `THREADS_APP_ID` and `THREADS_APP_SECRET`). Note that if `THREADS_APP_ID` and `THREADS_APP_SECRET` are missing at boot, Threads is hidden from the channel list entirely.
- **Match Exact Variable Names:** Postiz expects specific environment variable prefixes per platform in Configuration Reference. For example, X uses `X_API_KEY` and `X_API_SECRET` (not `TWITTER_CLIENT_ID`), while Facebook uses `FACEBOOK_APP_ID` (not `FACEBOOK_CLIENT_ID`).
- **Align OAuth Redirect URIs with FRONTEND_URL:** Postiz constructs all OAuth callback URLs using the pattern `{FRONTEND_URL}/integrations/social/{provider}`. Since your `.env` sets `FRONTEND_URL="http://localhost:4200"`, every developer portal must whitelist `http://localhost:4200/integrations/social/{provider}` without trailing slashes.
- **Restart Postiz After Editing .env:** Postiz reads environment variables only on boot. After updating `.env`, you must restart the backend and orchestrator processes (or recreate your Docker containers via `docker compose up -d --force-recreate`) for the new keys to take effect.

2. Master OAuth Provider & Redirect URI Reference

The table below summarizes the exact environment variables, provider identifiers, and local redirect URIs required for your `http://localhost:4200` setup (for standard Docker Compose on port 5000, replace `4200` with `5000`; for production, replace with `https://your-domain.com`).

Platform Channel	Postiz .env Variables	OAuth Redirect URI (Local Port 4200)	Key Portal Requirement
X (Twitter)	`X_API_KEY` `X_API_SECRET`	`http://localhost:4200/integrations/social/x`	App Type MUST be **Native App** (OAuth 1.0a Consumer Keys)

Platform Channel	Postiz .env Variables	OAuth Redirect URI (Local Port 4200)	Key Portal Requirement
LinkedIn (Profile)	`LINKEDIN_CLIENT_ID` `LINKEDIN_CLIENT_SECRET`	`http://localhost:4200/integrations/social/linkedin`	Requires **Advertising API**, Share on LinkedIn, & OpenID Connect
LinkedIn Page	`LINKEDIN_CLIENT_ID` `LINKEDIN_CLIENT_SECRET`	`http://localhost:4200/integrations/social/linkedin-page`	Shares the same LinkedIn App credentials; verify Company Page
Facebook Page	`FACEBOOK_APP_ID` `FACEBOOK_APP_SECRET`	`http://localhost:4200/integrations/social/facebook`	App Type: **Other → Business**; switch App Mode to **Live**
Instagram (FB Business)	`FACEBOOK_APP_ID` `FACEBOOK_APP_SECRET`	`http://localhost:4200/integrations/social/instagram`	Shares `FACEBOOK_APP_ID`; uses Facebook Login for Business
Instagram (Standalone)	`INSTAGRAM_APP_ID` `INSTAGRAM_APP_SECRET`	`http://localhost:4200/integrations/social/instagram-standalone`	Uses dedicated **Instagram App ID** under Instagram product settings
Threads	`THREADS_APP_ID` `THREADS_APP_SECRET`	`http://localhost:4200/integrations/social/threads`	Uses dedicated **Threads App ID**; accept Threads Tester invite
TikTok	`TIKTOK_CLIENT_ID` `TIKTOK_CLIENT_SECRET`	`https://localhost:4200/integrations/social/tiktok`	Requires **HTTPS** redirect URI, verified domain, & public media URL
Reddit	`REDDIT_CLIENT_ID` `REDDIT_CLIENT_SECRET`	`http://localhost:4200/integrations/social/reddit`	Select **web app** radio button at reddit.com/prefs/apps
Pinterest	`PINTEREST_CLIENT_ID` `PINTEREST_CLIENT_SECRET`	`http://localhost:4200/integrations/social/pinterest`	Requires a **Pinterest Business (Company) Account** & app approval

3. Step-by-Step Developer Portal Setup by Platform

3.1 X (Twitter) — Native App & OAuth 1.0a Consumer Keys

As documented in the Postiz X (Twitter) Guide, X's OAuth 2.0 flow only supports the v2 API, whereas uploading images requires the legacy v1.1 media endpoint. Therefore, Postiz uses the **OAuth 1.0a flow** (which supports both v1.1 and v2).

1. **Create Application:** Sign in to the X Developer Portal, create a Project and App (a Free tier account is sufficient for basic posting).
2. **Configure User Authentication Settings:** Open your App settings, scroll to **User authentication settings**, and click **Set up**.
3. **Set Permissions & App Type (Critical):**
   - Under **App permissions**, select **Read and write**.

- Under **Type of App**, select **Native App** (Public client). *Warning: Selecting "Web App, Automated App or Bot" will cause OAuth 1.0a authentication to fail with Error Code 32.*

4. **Set Callback URI:** Under **App info**, enter your Callback URI / Redirect URL:

```
Local:      http://localhost:4200/integrations/social/x
Docker:     http://localhost:5000/integrations/social/x
Production: https://your-postiz-domain.com/integrations/social/x
HTTPS Proxy (if X rejects http localhost): https://redirectmeto.com/http://localhost:4200/integrations/social/x
```

   Enter any valid URL (such as `https://example.com`) in the required **Website URL** field and click **Save**.

5. **Extract Consumer Keys:** Navigate to the **Keys and tokens** tab at the top of your App page. Under **Consumer Keys** (*not* the OAuth 2.0 Client ID and Client Secret below it), click **Regenerate** and copy the **API Key** and **API Key Secret** into `X_API_KEY` and `X_API_SECRET`.

3.2 LinkedIn & LinkedIn Page — Shared App with Advertising API

Both personal profiles (LinkedIn) and organization pages (LinkedIn Page) share the exact same `LINKEDIN_CLIENT_ID` and `LINKEDIN_CLIENT_SECRET` variables.

1. **Create & Verify App:** Visit the LinkedIn Developer Portal and click **Create app**. Associate it with your LinkedIn Company Page and complete the verification link under the **Settings** tab.
2. **Enable All Three Required Products:** Open the **Products** tab and request access to all three of these products:
   - **Share on LinkedIn** (grants `w_member_social`)
   - **Sign In with LinkedIn using OpenID Connect** (grants `openid`, `profile`)
   - **Advertising API** (grants `r_basicprofile`, `rw_organization_admin`, `w_organization_social`, `r_organization_social`)

   Why Advertising API is mandatory: Even if you only connect a personal LinkedIn profile, Postiz requests all 7 scopes during login and relies on the Advertising API for token refreshes. If any scope is missing from your app's **Auth** tab, LinkedIn silently omits it from the token and Postiz rejects the login with a `NotEnoughScopes` error.
3. **Verify Scopes & Redirect URIs:** Open the **Auth** tab. Confirm that the **OAuth 2.0 scopes** box lists all 7 scopes (`openid`, `profile`, `w_member_social`, `r_basicprofile`, `rw_organization_admin`, `w_organization_social`, `r_organization_social`). Under **OAuth 2.0 settings → Authorized redirect URLs for your app**, register **both** endpoints:

```
http://localhost:4200/integrations/social/linkedin
http://localhost:4200/integrations/social/linkedin-page
```

4. **Copy Credentials:** Under the **Auth** tab (Application credentials), copy **Client ID** and **Primary Client Secret** into `LINKEDIN_CLIENT_ID` and `LINKEDIN_CLIENT_SECRET`.

3.3 Facebook (Page/Business) & Instagram (Facebook Business)

Per the Postiz Facebook Guide and Instagram Guide, Facebook Pages and Instagram accounts linked to a Facebook Business Page share a single Meta application and the same `FACEBOOK_APP_ID` and `FACEBOOK_APP_SECRET` credentials.

1. **Create Business App:** Go to Meta for Developers and click **Create App**. Select your Business Portfolio (requires 2FA enabled on your Meta account).
2. **Select App Type:** On the "Add use cases" screen, click the **All** filter on the left, scroll to the bottom under *"Looking for something else?"*, and choose **Other** → **Next**. Select **Business** as the app type and complete creation.
3. **Configure Facebook Login for Business:** From the App Dashboard, click **Set up** on **Facebook Login for Business**. In the left sidebar under *Facebook Login for Business → Settings*, enable **Client OAuth Login** and **Web OAuth Login**, then add both Redirect URIs to **Valid OAuth Redirect URIs**:

```
http://localhost:4200/integrations/social/facebook
http://localhost:4200/integrations/social/instagram
```

4. **Request Permissions (App Review):** Under *App Review → Permissions and Features*, request access for the following scopes (note: if your install is strictly for personal use by accounts listed under *App Roles → Admins/Developers/Testers*, standard access works immediately without formal Business Verification):

- **Facebook Scopes:** `pages_show_list`, `business_management`, `pages_manage_posts`, `pages_manage_engagement`, `pages_read_engagement`, `read_insights`
- **Instagram (FB Business) Scopes:** `instagram_basic`, `instagram_content_publish`, `instagram_manage_comments`, `instagram_manage_insights` (plus `pages_show_list`, `pages_read_engagement`, `business_management`)

5. **Switch App Mode to Live:** At the top header bar of the Meta dashboard, toggle **App Mode** from **Development** to **Live** (requires adding a Privacy Policy URL in *App Settings → Basic*). If left in Development mode, posts published via Postiz will appear without images or remain invisible to the public.
6. **Copy Main App Credentials:** Navigate to *App Settings → Basic* in the left sidebar. Copy the top-level **App ID** and **App Secret** into `FACEBOOK_APP_ID` and `FACEBOOK_APP_SECRET`.

3.4 Instagram (Standalone Business Login)

If your Instagram Professional (Creator or Business) account is *not* linked to a Facebook Page, use the **Instagram (Standalone)** channel in Postiz (Instagram Standalone Setup).

1. **Add Instagram Product:** Inside your Meta Business App dashboard, scroll to **Add products to your app** and click **Set up** on **Instagram**.
2. **Configure Business Login:** Navigate to *Instagram → API setup with Instagram business login* and click **Set up Instagram business login**.
3. **Add Standalone Redirect URI:** In the OAuth Redirect URI configuration box, enter:

   ```
   http://localhost:4200/integrations/social/instagram-standalone
   ```

4. **Copy Dedicated Instagram App ID (Prevents "Invalid App ID"):** On the *API setup with Instagram business login* screen, Meta displays a dedicated **Instagram App ID** and **Instagram App Secret** that are completely different from your main Facebook App ID. Copy these specific values into `INSTAGRAM_APP_ID` and `INSTAGRAM_APP_SECRET`.
5. **Add & Accept Instagram Tester Role:** To avoid the `Insufficient developer role` error before public App Review, go to *App Roles → Roles → Add People*, choose **Instagram Tester**, and enter your Instagram handle. Then open Instagram on web or mobile, go to *Settings → Website Permissions → Apps and Websites → Tester Invites*, and click **Accept**.

3.5 Threads — Dedicated Use Case App ID & Tester Handshake

As detailed in the Postiz Threads Guide, Threads uses Meta's newer Use Cases dashboard and requires its own App ID and Secret.

1. **Create Threads App:** In Meta for Developers, click **Create App** and select the **Access the Threads API** use case. (If Meta prompts *"There are no business portfolios available"*, ensure Two-Factor Authentication is enabled in your Meta Business Suite settings).
2. **Add Threads Scopes:** On the App Dashboard, click **Access the Threads API** → **Permissions**, and ensure both `threads_basic` and `threads_content_publish` are added.
3. **Set Redirect & Callback URLs:** Click the **Settings** tab inside the *Access the Threads API* section. Enter the Redirect URI in **Redirect Callback URLs**, and also populate **Uninstall Callback URL** and **Delete Callback URL** with the same URL (Meta will not allow the form to save if those two fields are blank). **Crucial:** After typing each URL, you must click the dropdown/pill suggestion that appears beneath the input box to make the URL active before saving:

   ```
   http://localhost:4200/integrations/social/threads
   ```

4. **Copy Threads App ID & Secret:** Still on the *Access the Threads API → Settings* tab, copy the **Threads App ID** (*not* the top-level App ID in Basic Settings) and click **Show** next to **Threads App Secret**. Scroll all the way across the narrow text box to copy the full 32-character secret. Assign these to `THREADS_APP_ID` and `THREADS_APP_SECRET`.
5. **Finish Wizard & Accept Tester Invite:** Return to the main Dashboard view, click **Finish customization**, and select **Yes I'm finished**. Next, go to *App Roles → Roles → Testers → Add People*, select **Threads Tester**, and add your Threads username. Finally, visit threads.com → *Settings → Account → Website permissions → Invites* and accept the pending invite.

3.6 TikTok — HTTPS Requirement, Login Kit & Content Posting API

Per the Postiz TikTok Guide, TikTok enforces strict HTTPS and domain-verification rules that affect local self-hosted setups:

1. **Understand TikTok's HTTPS & Media Constraints:**
   - **HTTPS Redirect URI Only:** TikTok's developer console rejects plain `http://` redirect URIs (even for `localhost`). You must register `https://localhost:4200/integrations/social/tiktok` or expose Postiz via an HTTPS tunnel/reverse proxy (such as Caddy, Cloudflare Tunnel, or ngrok) and set `FRONTEND_URL` accordingly when connecting TikTok.
   - **Public Media Fetching (`pull_from_url`):** When publishing videos or photos, TikTok fetches media from your server via URL. Local filesystem paths (`STORAGE_PROVIDER="local"` on `localhost`) cannot be reached by TikTok's servers. You must either configure Cloudflare R2 (`STORAGE_PROVIDER="cloudflare"`) or expose your uploads over a verified HTTPS domain.
   - **Unaudited App Behavior:** Until your TikTok app passes TikTok's formal compliance audit, Direct Post usage is capped at 5 users per 24 hours, test accounts must be set to Private, and all posts are forced to `SELF_ONLY` (private) visibility.
2. **Create App & Verify Domain:** Go to the TikTok Developer Portal, create a new app, tick **Web** under platforms, and provide HTTPS URLs for your Terms of Service and Privacy Policy on a public domain you own and have verified with TikTok.
3. **Add Products & Redirect URI:** Under **Products**, add both **Login Kit** and **Content Posting API**. Inside *Login Kit*, set the Redirect URI to `https://localhost:4200/integrations/social/tiktok` (or your HTTPS tunnel URL). Inside *Content Posting API*, toggle **Direct Post** ON.
4. **Add Required Scopes:** Under Scopes, enable all 5 permissions:

   ```
   user.info.basic, user.info.profile, video.create, video.publish, video.upload
   ```

5. **Copy Client Key & Secret:** Copy the 16-character **Client Key** into `TIKTOK_CLIENT_ID` and the 32-character **Client Secret** into `TIKTOK_CLIENT_SECRET`.

3.7 Reddit — Web App Configuration

Follow the Postiz Reddit Guide to register a standard OAuth2 web application:

1. **Create Reddit App:** Navigate to Reddit Authorized Applications (reddit.com/prefs/apps), scroll to the bottom, and click **create another app...** (or **are you a developer? create an app...**).
2. **Configure App Type & Redirect URI:**
   - **Name:** `MyPostizInstance` (or any descriptive label)
   - **App Type Radio Button:** Select **web app** (do *not* select "installed app" or "script").
   - **redirect uri:** Enter `http://localhost:4200/integrations/social/reddit`

   Click **create app**.
3. **Extract Client ID & Secret:** Copy the 14-character string displayed directly underneath the *"web app"* heading (top-left under your app name) as `REDDIT_CLIENT_ID`, and copy the `secret` field as `REDDIT_CLIENT_SECRET`.

3.8 Pinterest — Business Account & Developer App

Follow the Postiz Pinterest Guide to configure Pinterest:

1. **Convert or Create a Business Account:** Pinterest API access requires a free **Pinterest Business (Company) Account**. Log in to Pinterest and convert your account or create a new Business profile.
2. **Create & Submit Developer App:** Open the Pinterest Developer Dashboard, click **Connect app** / **Create app**, complete the required application details, and wait for Trial/Standard access approval.
3. **Configure Redirect URI:** Open your Pinterest App page, locate the **Redirect URIs** box, and add:

   ```
   http://localhost:4200/integrations/social/pinterest
   ```

4. **Copy Credentials:** Copy the numeric **App id** into `PINTEREST_CLIENT_ID` and the **App secret key** into `PINTEREST_CLIENT_SECRET`.

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4. Complete Postiz .env File with Social Channel Credentials

Append the **Social Provider Keys** section below directly to your existing `.env` file. Ensure there are no spaces around the = sign, no trailing spaces inside quotes, and that each credential matches the exact portal field described in Section 3:

```
# Configuration reference: https://docs.postiz.com/self-host/configuration/reference

# === Required Settings
DATABASE_URL="postgresql://postiz-user:postiz-password@localhost:5432/postiz-db-local"
REDIS_URL="redis://localhost:6379"
JWT_SECRET="random string for your JWT secret, make it long"

# === This needs to be exactly the URL you're accessing Postiz on
FRONTEND_URL="http://localhost:4200"
NEXT_PUBLIC_BACKEND_URL="http://localhost:3000"
BACKEND_INTERNAL_URL="http://localhost:3000"

# === Storage Configuration
# Note: TikTok requires publicly reachable HTTPS media URLs (e.g. Cloudflare R2).
STORAGE_PROVIDER="local"
UPLOAD_DIRECTORY="/path/to/your/local/uploads"
NEXT_PUBLIC_UPLOAD_STATIC_DIRECTORY="/uploads"

CLOUDFLARE_ACCOUNT_ID="your-account-id"
CLOUDFLARE_ACCESS_KEY="your-access-key"
CLOUDFLARE_SECRET_ACCESS_KEY="your-secret-access-key"
CLOUDFLARE_BUCKETNAME="your-bucket-name"
CLOUDFLARE_BUCKET_URL="https://your-bucket-url.r2.cloudflarestorage.com/"
CLOUDFLARE_REGION="auto"

# === Social Provider Credentials (Add Below) ===

# 1. X (Twitter) — Consumer Keys (OAuth 1.0a) from "Native App" setup
X_API_KEY="your-x-consumer-api-key"
X_API_SECRET="your-x-consumer-api-secret"
# Optional: set to true if on Free X tier without analytics access
DISABLE_X_ANALYTICS="true"

# 2. LinkedIn (Personal Profile & LinkedIn Company Page share these keys)
LINKEDIN_CLIENT_ID="your-linkedin-client-id"
LINKEDIN_CLIENT_SECRET="your-linkedin-client-secret"

# 3. Facebook Page & Instagram (Facebook Business) — Main Meta App ID & Secret
# Found under Meta App Dashboard -> App Settings -> Basic
FACEBOOK_APP_ID="your-main-meta-app-id"
FACEBOOK_APP_SECRET="your-main-meta-app-secret"

# 4. Instagram (Standalone) — Dedicated Instagram App ID & Secret
# Found under Meta App Dashboard -> Instagram -> API setup with business login
INSTAGRAM_APP_ID="your-dedicated-instagram-app-id"
INSTAGRAM_APP_SECRET="your-dedicated-instagram-app-secret"

# 5. Threads — Dedicated Threads App ID & 32-char Secret
# Found under Meta App Dashboard -> Access the Threads API -> Settings
THREADS_APP_ID="your-dedicated-threads-app-id"
THREADS_APP_SECRET="your-32-character-threads-app-secret"

# 6. TikTok — 16-char Client Key & 32-char Client Secret
TIKTOK_CLIENT_ID="your-16-char-tiktok-client-key"
TIKTOK_CLIENT_SECRET="your-32-char-tiktok-client-secret"

# 7. Reddit — 14-char Web App ID & Secret
REDDIT_CLIENT_ID="your-reddit-web-app-client-id"
REDDIT_CLIENT_SECRET="your-reddit-web-app-secret"

# 8. Pinterest — App ID & App Secret Key
PINTEREST_CLIENT_ID="your-pinterest-app-id"
PINTEREST_CLIENT_SECRET="your-pinterest-app-secret-key"
```

5. Post-Configuration Verification & Troubleshooting Checklist

If a social channel still fails to connect after populating the variables above, verify these common self-hosted failure points documented in OAuth & Channel Connect Errors and Self-Host Gotchas:

Symptom / Error Message	Root Cause	Resolution
"Invalid App ID" (Facebook, Instagram, Threads)	Empty `.env` variable, un-restarted backend container, or using the top-level Meta App ID for Instagram Standalone / Threads.	Verify you copied the sub-product App ID from *Instagram → API setup* or *Threads API → Settings*, then restart Postiz so the backend reloads `.env`.
"Invalid state" flash error after OAuth redirect	Session CSRF state token cookie was dropped or mismatched between `FRONTEND_URL` and `NEXT_PUBLIC_BACKEND_URL`.	Access Postiz strictly via `http://localhost:4200` (never mix `127.0.0.1` and `localhost`), disable Brave Shields / ad-blockers, and avoid incognito windows.
"invalid_grant" during token exchange	Redirect URI mismatch (including a trailing slash) or re-using an expired single-use OAuth authorization code.	Confirm the developer portal Redirect URI matches `http://localhost:4200/integrations/social/{provider}` character-for-character and start a fresh connect from Postiz.
"Error 32: Could not authenticate you" on X	X App Type was set to "Web App, Automated App or Bot" or OAuth 2.0 keys were used instead of Consumer Keys.	Change X App Type to **Native App** and copy the **Consumer Keys** (API Key & Secret) into `X_API_KEY` and `X_API_SECRET`.
"NotEnoughScopes" on LinkedIn	Missing the **Advertising API** product on your LinkedIn Developer App.	Request **Advertising API** under the *Products* tab and verify all 7 scopes appear under the *Auth* tab.
"Insufficient developer role" on Instagram / Threads	Target account is not registered as a Tester while the Meta app is unreviewed.	Add the handle as an **Instagram Tester** or **Threads Tester** under *App Roles → Roles*, then accept the invite inside Instagram/Threads settings.